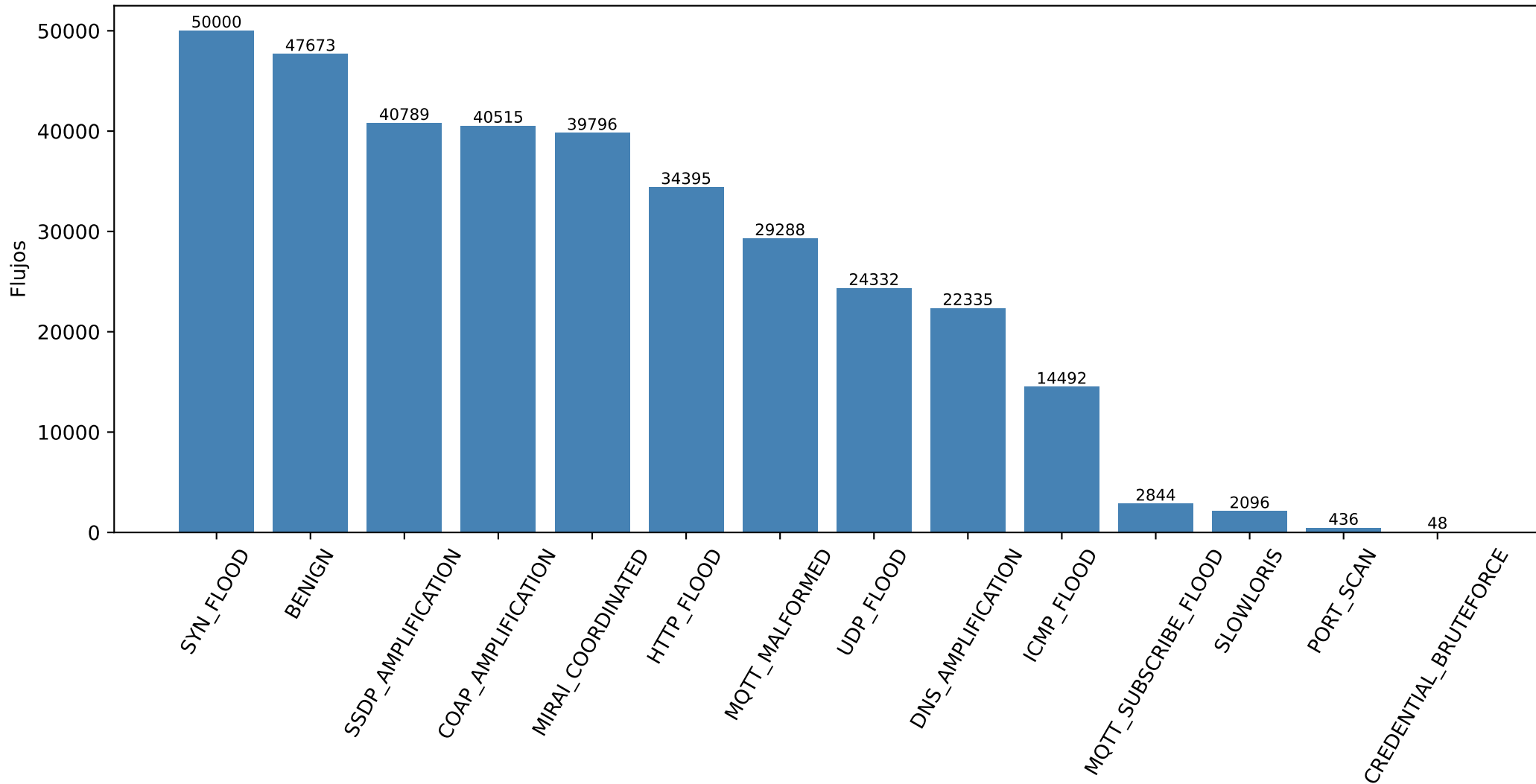


# IoT-SDN-AI — Reporte de evaluación

run: run\_20260924-192239

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| Dataset:    | 449,966 flujos                                                                           |
| Features:   | 50                                                                                       |
| Clases:     | 14 (BENIGN, COAP_AMPLIFICATION, CREDENTIAL_BRUTEFORCE, DNS_AMPLIFICATION, HTTP_FLOOD...) |
| Modelos:    | RandomForest, XGBoost, MLP, IsolationForest, AutoEncoder                                 |
| Estrategia: | grouped split anti-fuga (StratifiedGroupKFold fold0) 80/20, StandardScaler               |

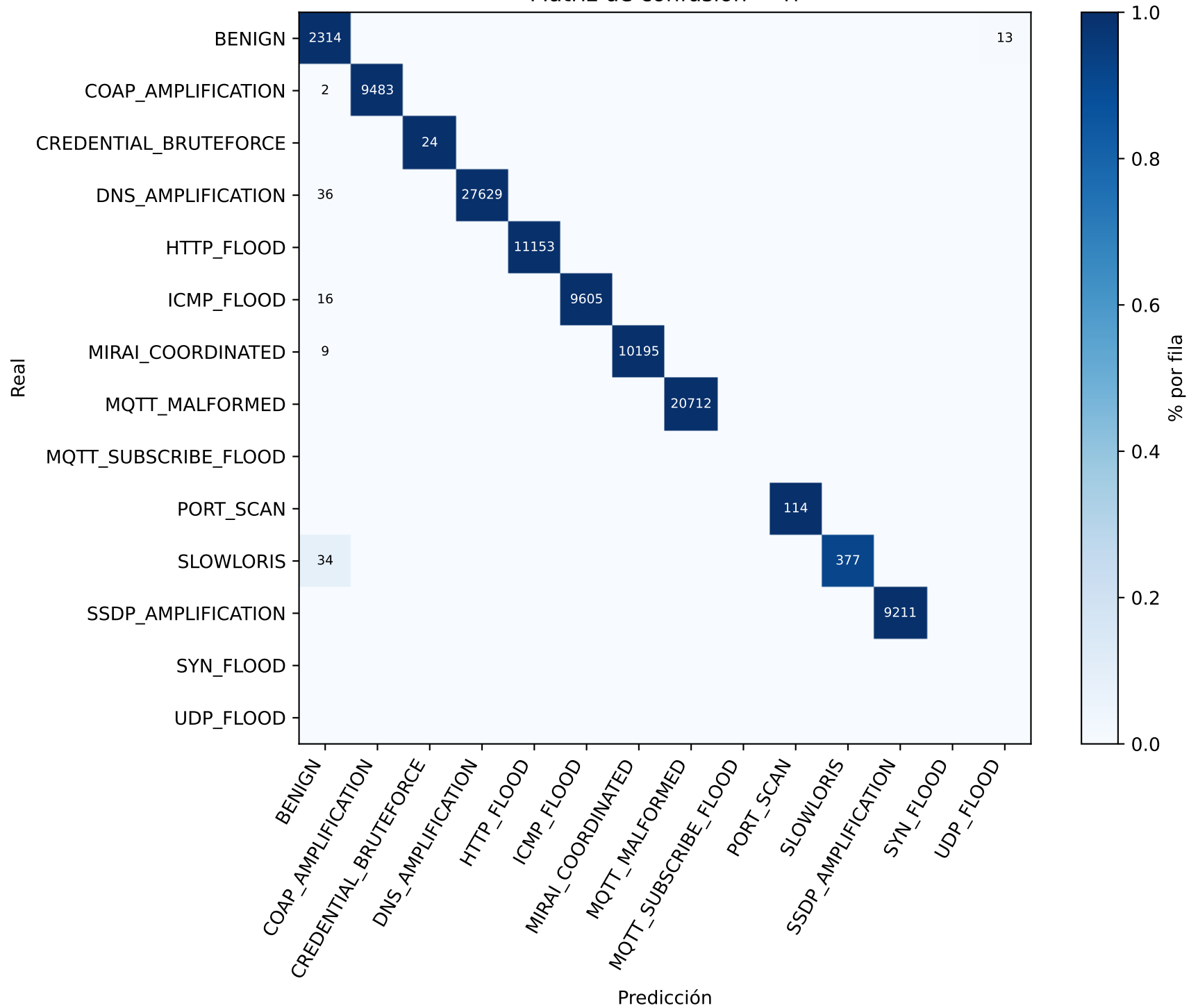
Distribución de clases en el dataset



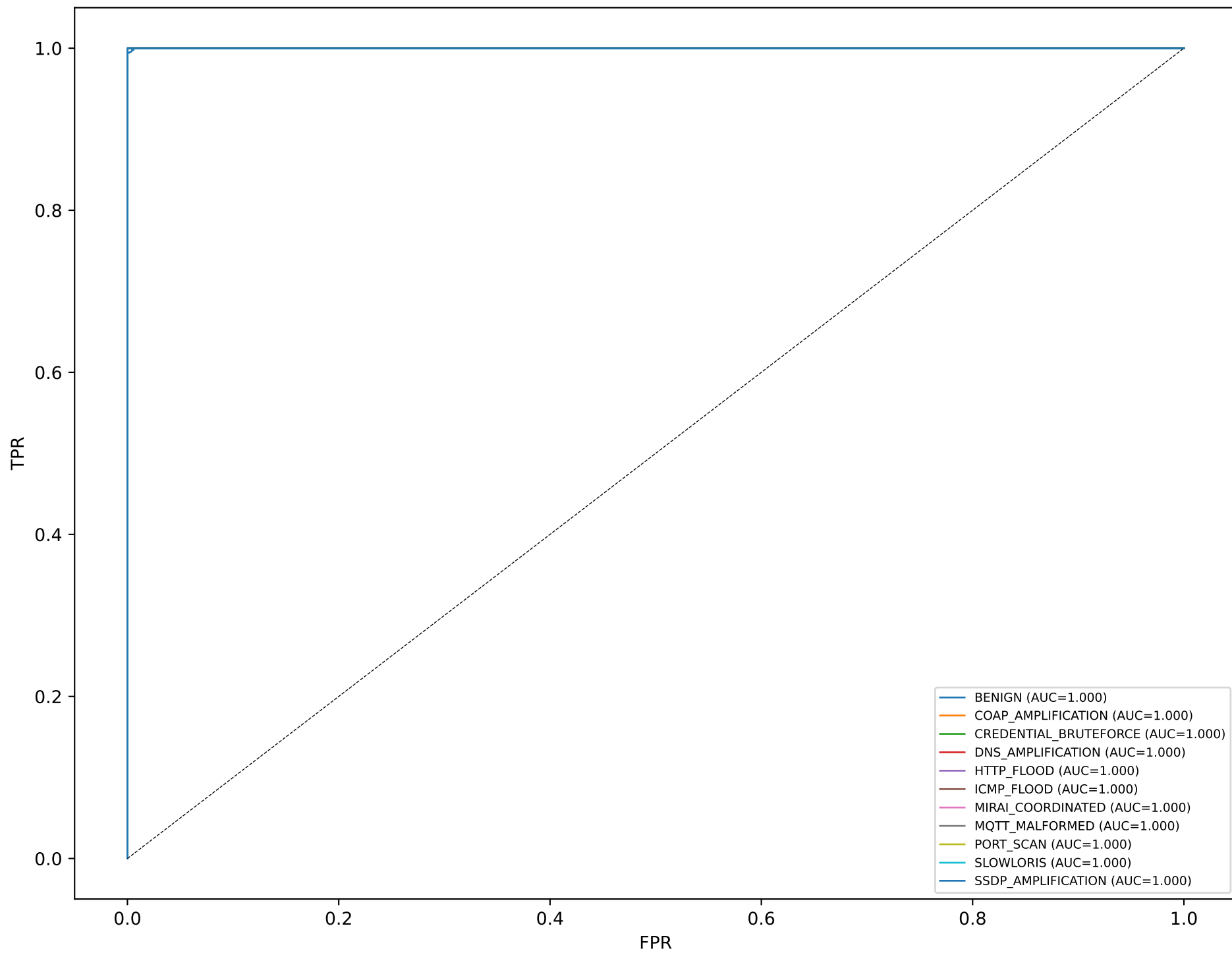
Classification report — rf

|                       | precision | recall | f1-score | support |
|-----------------------|-----------|--------|----------|---------|
| BENIGN                | 0.96      | 0.99   | 0.98     | 2327    |
| COAP_AMPLIFICATION    | 1.00      | 1.00   | 1.00     | 9485    |
| CREDENTIAL_BRUTEFORCE | 1.00      | 1.00   | 1.00     | 24      |
| DNS_AMPLIFICATION     | 1.00      | 1.00   | 1.00     | 27665   |
| HTTP_FLOOD            | 1.00      | 1.00   | 1.00     | 11153   |
| ICMP_FLOOD            | 1.00      | 1.00   | 1.00     | 9621    |
| MIRAI_COORDINATED     | 1.00      | 1.00   | 1.00     | 10204   |
| MQTT_MALFORMED        | 1.00      | 1.00   | 1.00     | 20712   |
| PORT_SCAN             | 1.00      | 1.00   | 1.00     | 114     |
| SLOWLORIS             | 1.00      | 0.92   | 0.96     | 411     |
| SSDP_AMPLIFICATION    | 1.00      | 1.00   | 1.00     | 9211    |
| UDP_FLOOD             | 0.00      | 0.00   | 0.00     | 0       |
| accuracy              |           |        | 1.00     | 100927  |
| macro avg             | 0.91      | 0.91   | 0.91     | 100927  |
| weighted avg          | 1.00      | 1.00   | 1.00     | 100927  |

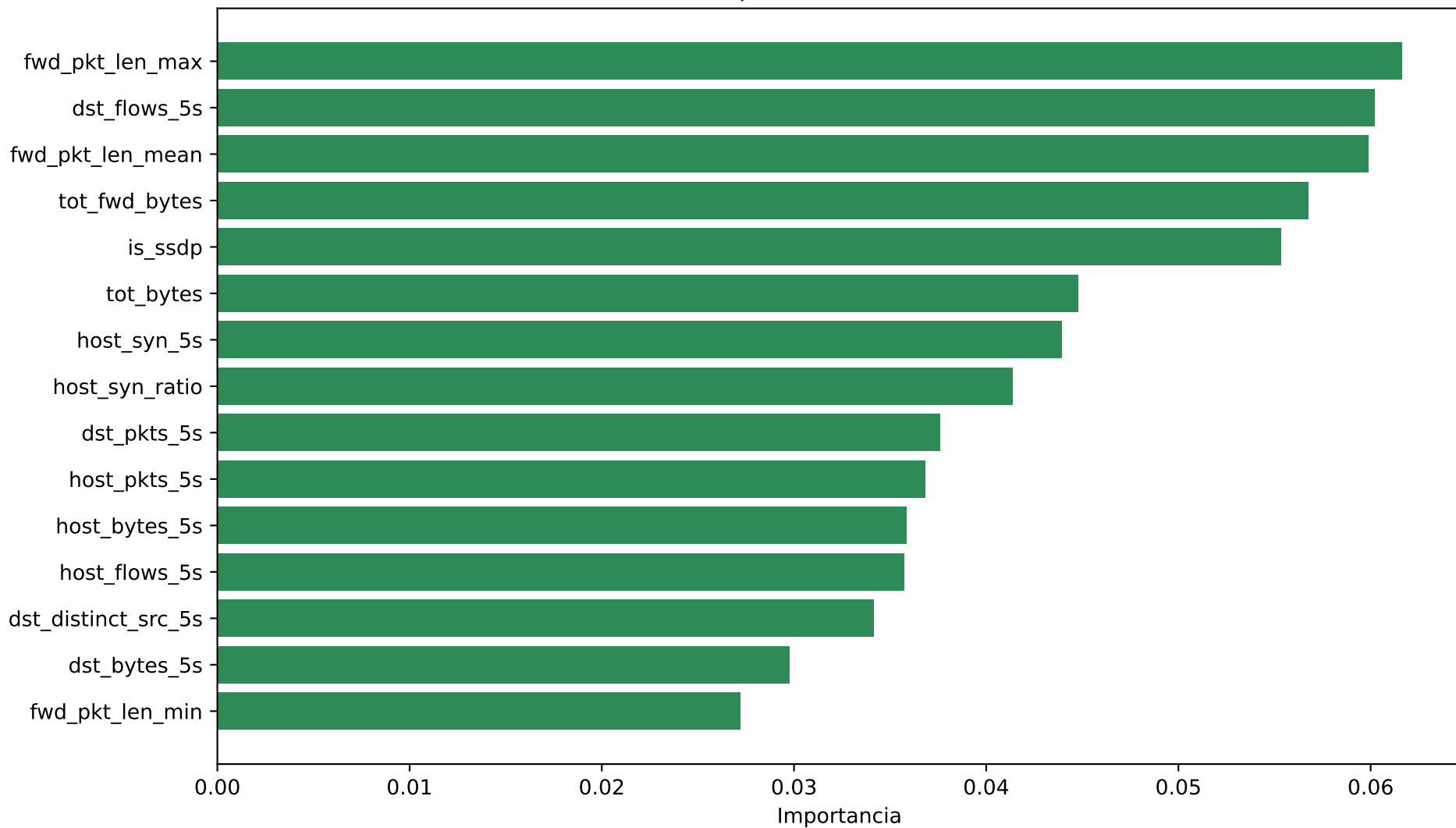
Matriz de confusión — rf



ROC OvR — rf



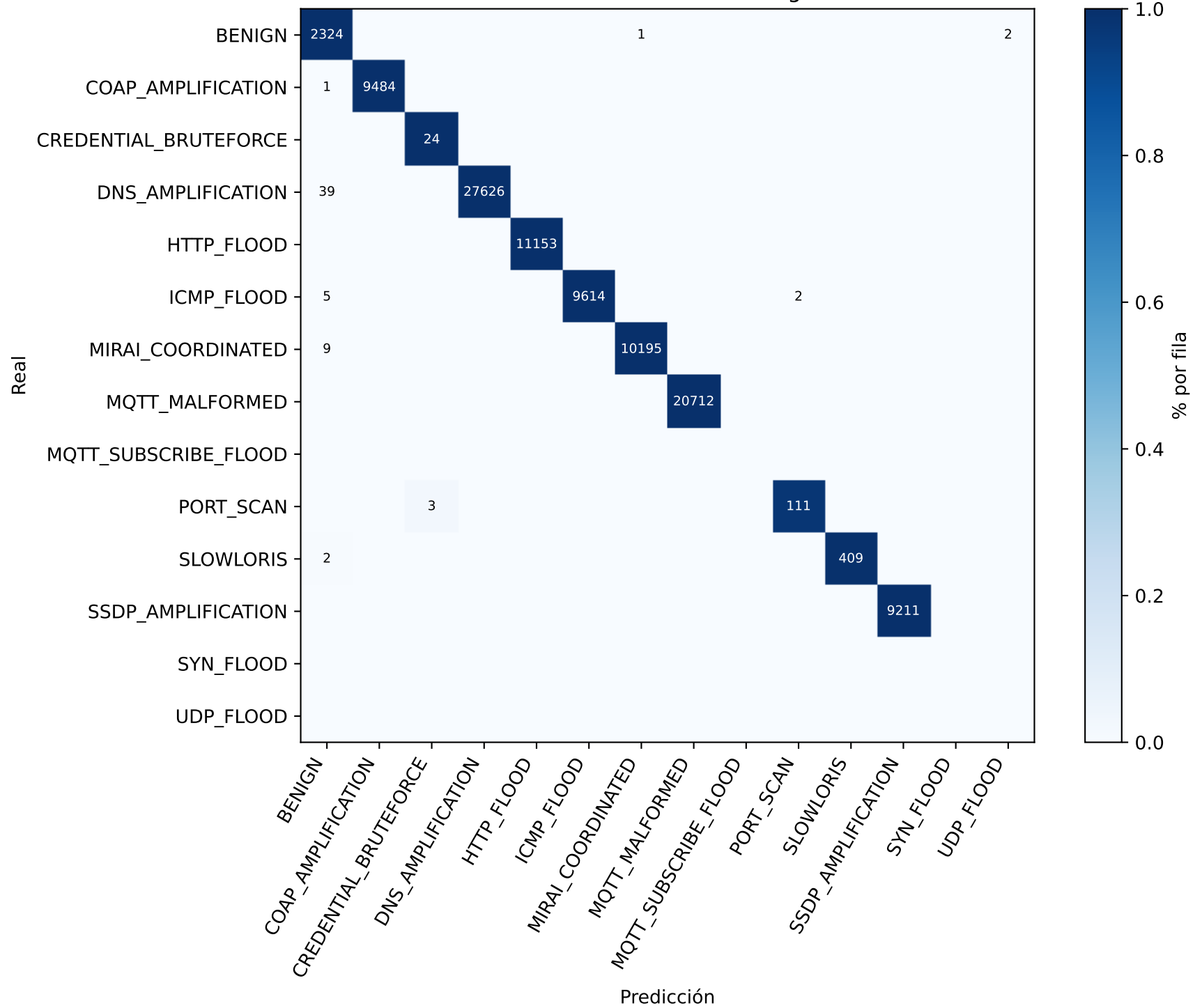
Top-15 features — rf



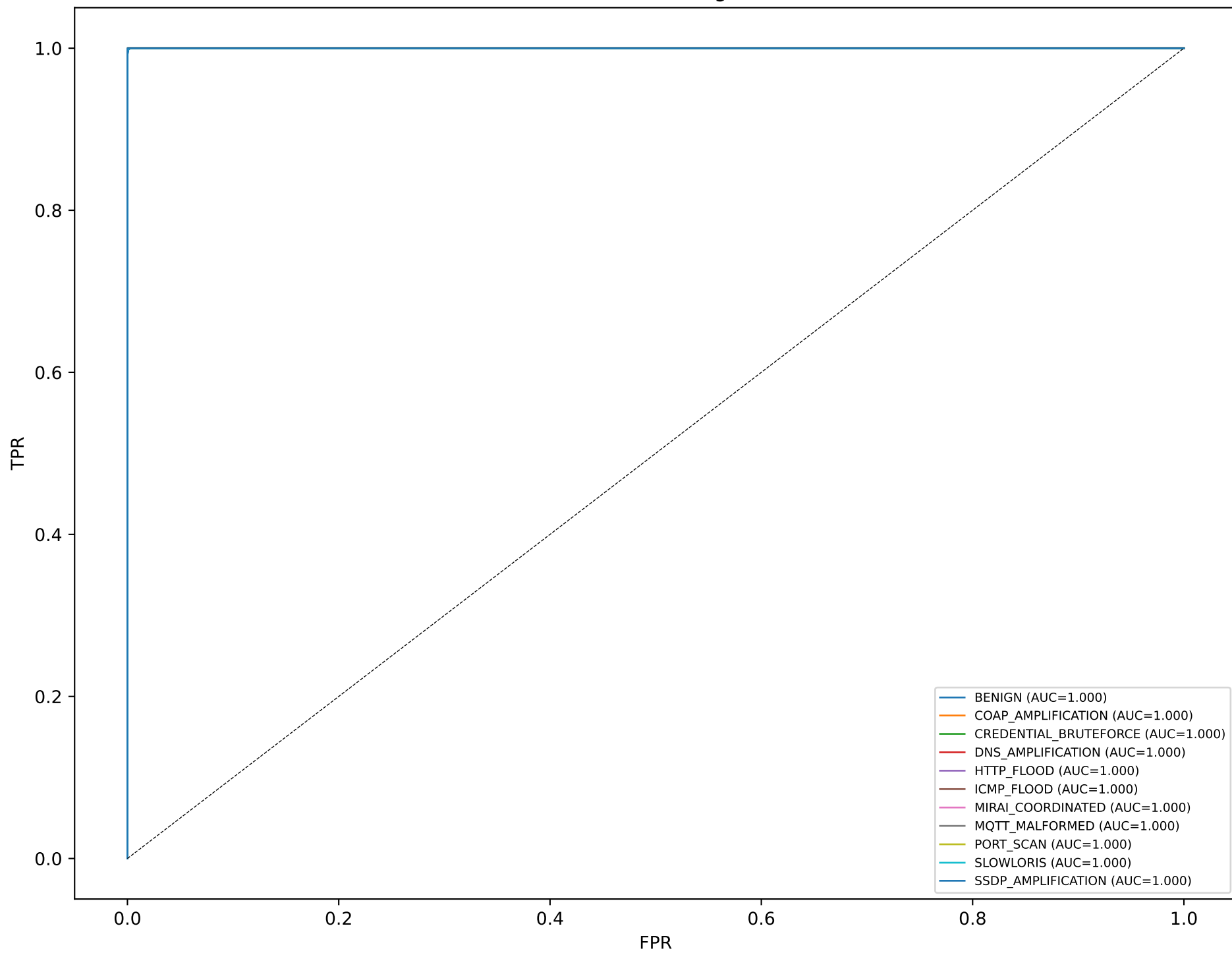
# Classification report — xgb

|                       | precision | recall | f1-score | support |
|-----------------------|-----------|--------|----------|---------|
| BENIGN                | 0.98      | 1.00   | 0.99     | 2327    |
| COAP_AMPLIFICATION    | 1.00      | 1.00   | 1.00     | 9485    |
| CREDENTIAL_BRUTEFORCE | 0.89      | 1.00   | 0.94     | 24      |
| DNS_AMPLIFICATION     | 1.00      | 1.00   | 1.00     | 27665   |
| HTTP_FLOOD            | 1.00      | 1.00   | 1.00     | 11153   |
| ICMP_FLOOD            | 1.00      | 1.00   | 1.00     | 9621    |
| MIRAI_COORDINATED     | 1.00      | 1.00   | 1.00     | 10204   |
| MQTT_MALFORMED        | 1.00      | 1.00   | 1.00     | 20712   |
| PORT_SCAN             | 0.98      | 0.97   | 0.98     | 114     |
| SLOWLORIS             | 1.00      | 1.00   | 1.00     | 411     |
| SSDP_AMPLIFICATION    | 1.00      | 1.00   | 1.00     | 9211    |
| UDP_FLOOD             | 0.00      | 0.00   | 0.00     | 0       |
| accuracy              |           |        | 1.00     | 100927  |
| macro avg             | 0.90      | 0.91   | 0.91     | 100927  |
| weighted avg          | 1.00      | 1.00   | 1.00     | 100927  |

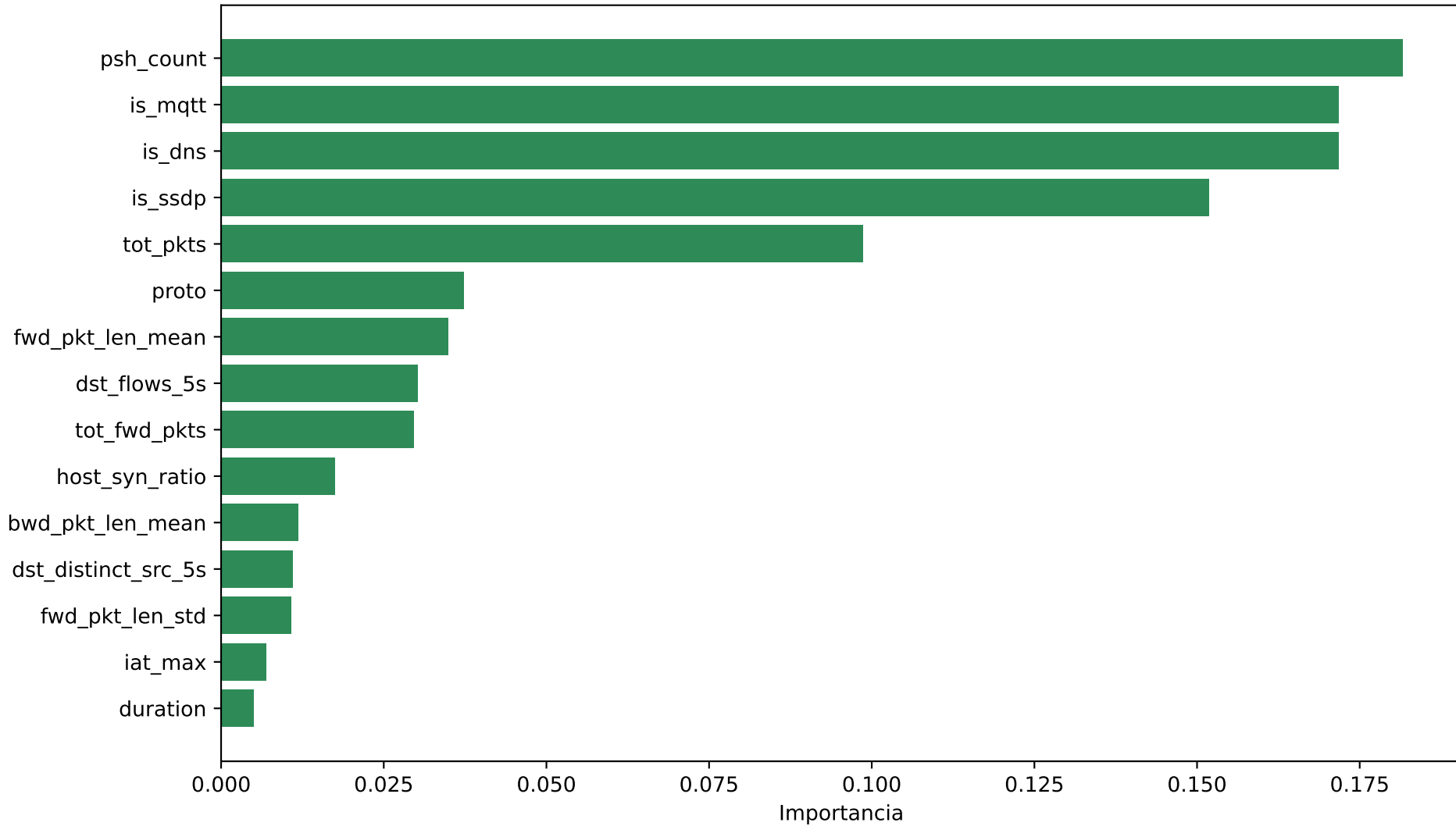
## Matriz de confusión — xgb



ROC OvR — xgb



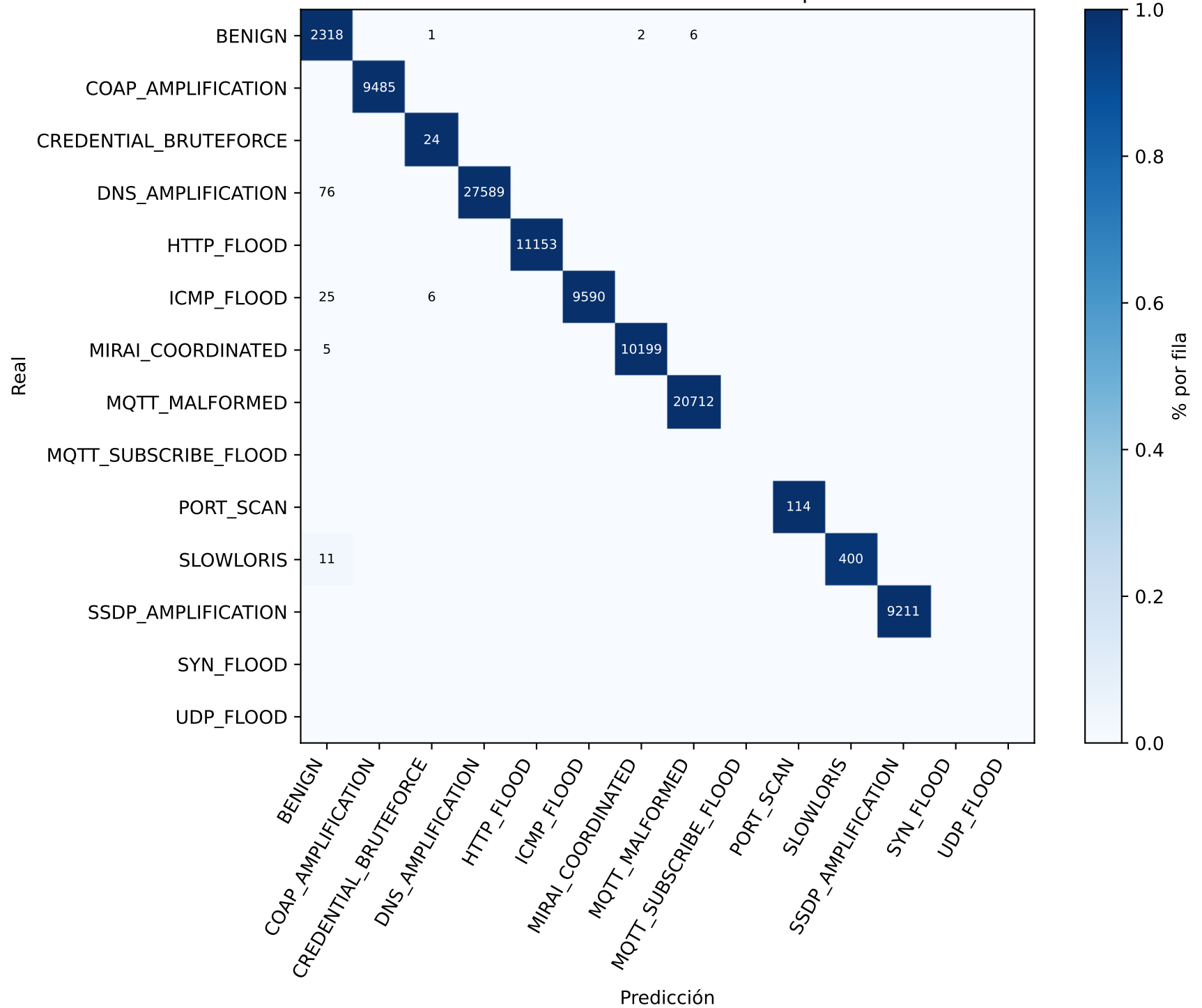
Top-15 features — xgb



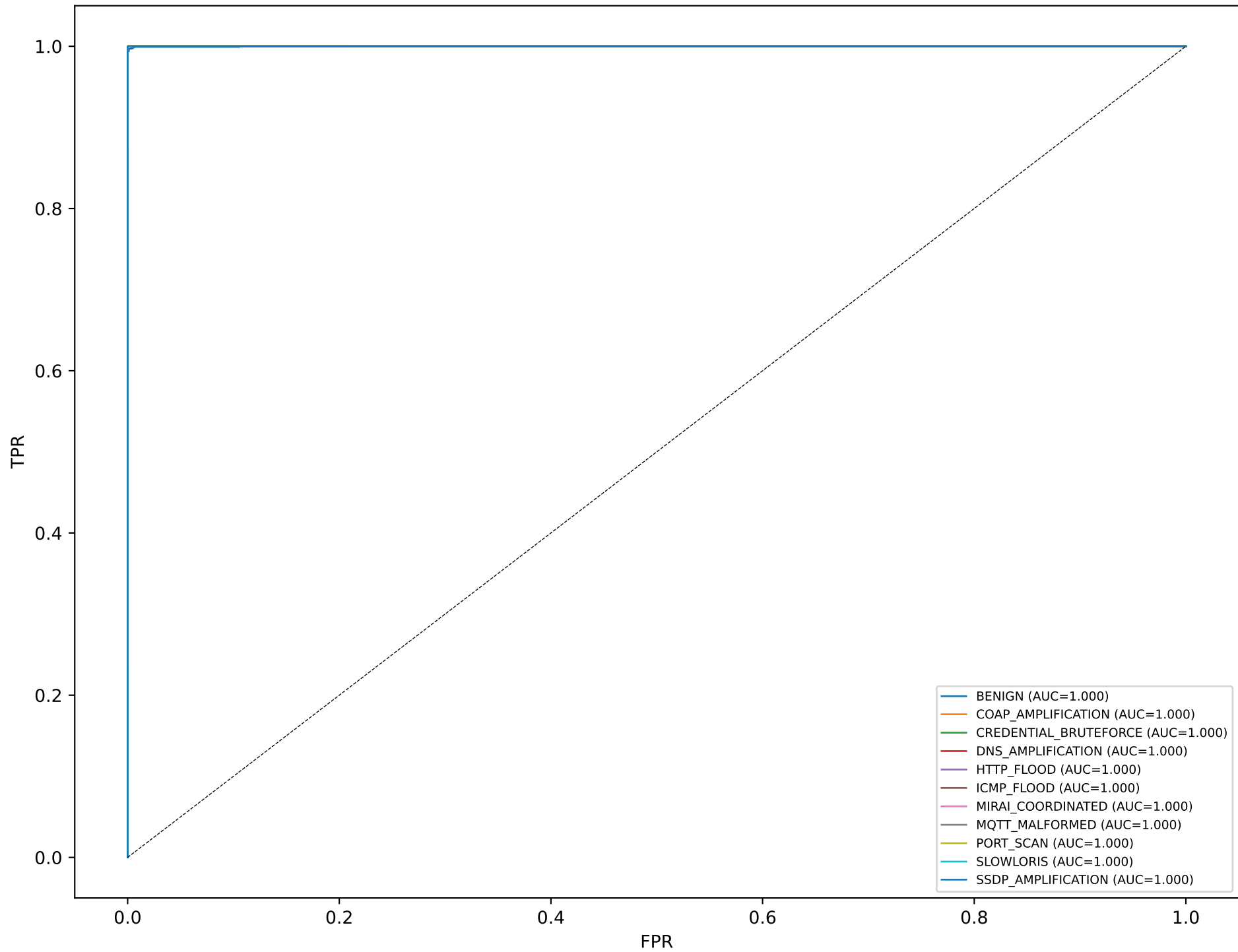
# Classification report — mlp

|                       | precision | recall | f1-score | support |
|-----------------------|-----------|--------|----------|---------|
| BENIGN                | 0.95      | 1.00   | 0.97     | 2327    |
| COAP_AMPLIFICATION    | 1.00      | 1.00   | 1.00     | 9485    |
| CREDENTIAL_BRUTEFORCE | 0.77      | 1.00   | 0.87     | 24      |
| DNS_AMPLIFICATION     | 1.00      | 1.00   | 1.00     | 27665   |
| HTTP_FLOOD            | 1.00      | 1.00   | 1.00     | 11153   |
| ICMP_FLOOD            | 1.00      | 1.00   | 1.00     | 9621    |
| MIRAI_COORDINATED     | 1.00      | 1.00   | 1.00     | 10204   |
| MQTT_MALFORMED        | 1.00      | 1.00   | 1.00     | 20712   |
| PORT_SCAN             | 1.00      | 1.00   | 1.00     | 114     |
| SLOWLORIS             | 1.00      | 0.97   | 0.99     | 411     |
| SSDP_AMPLIFICATION    | 1.00      | 1.00   | 1.00     | 9211    |
| accuracy              |           |        | 1.00     | 100927  |
| macro avg             | 0.98      | 1.00   | 0.98     | 100927  |
| weighted avg          | 1.00      | 1.00   | 1.00     | 100927  |

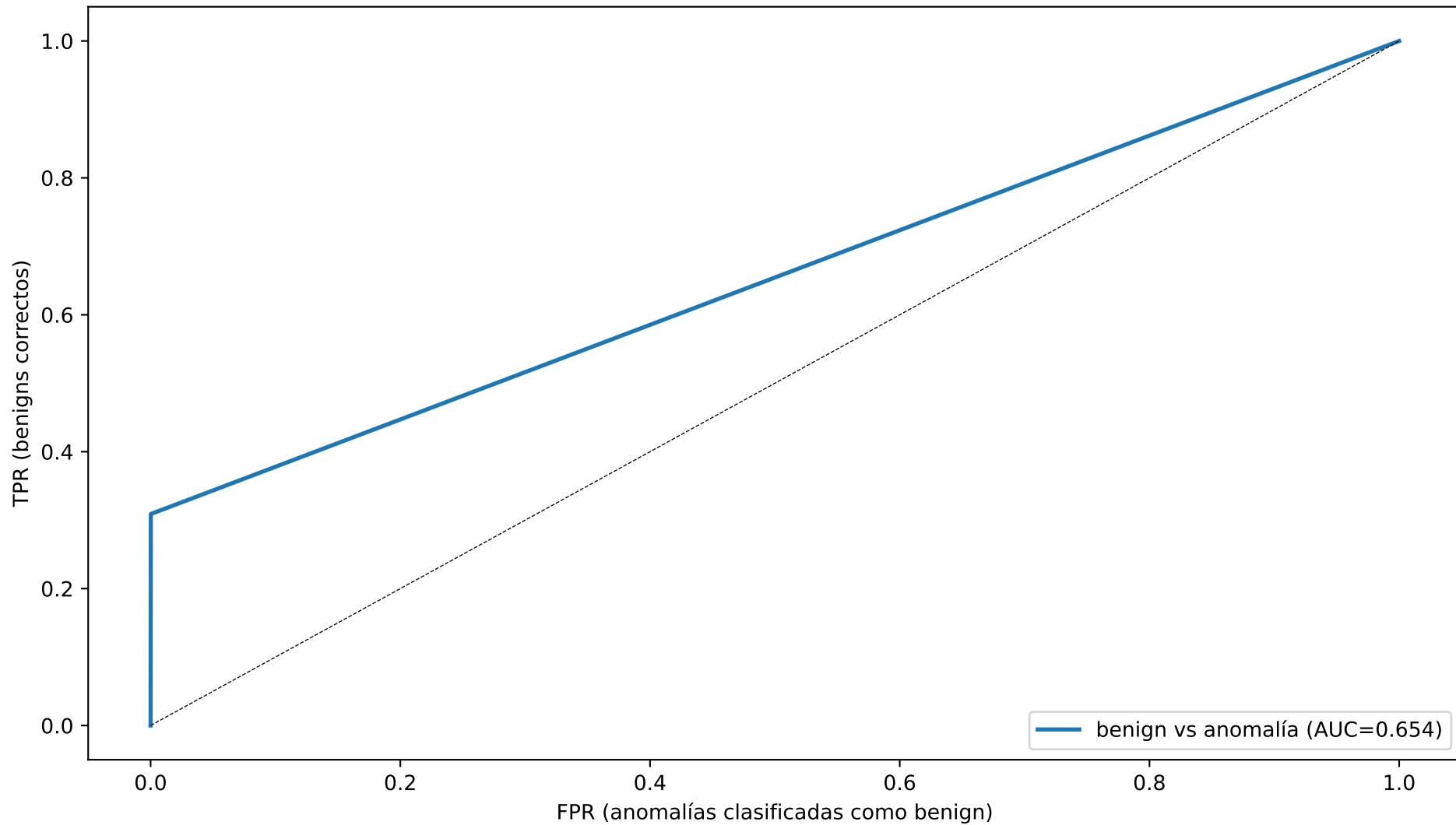
Matriz de confusión — mlp



ROC OvR — mlp



ROC binaria — isoforest



ROC binaria — autoencoder

